March 2026

Tutorial: Altimetry / DEM Module (QGIS)

Design: Roland (Ypsos)

1. Introduction

Module designed and specified by Roland (Ypsos) for Ortho4XP V3. It replaces the manual QGIS procedure (loading + extent spreadsheet + assembly) with a button interface. Everything runs through rasterio, with no Terminal command. Source files are never modified.

Overview

The Altimetry / DEM module automatically prepares and assembles the digital elevation model (DEM) of an Ortho4XP tile, from elevation data coming from several countries and providers.

Logical order of the steps (shown at the top of the window):

Create structure → Load the data → Prepare data (once per country) → Assemble tile

Key principle: assembly selects files by their GEOGRAPHIC EXTENT, never by folder name. Organising by country is only for YOUR convenience.

The 4-folder structure

The module works inside an « Altimetry » structure containing four folders, each with its own button:

- Source elevation / <Country> — your RAW provider data (French IGN, and Swiss, Belgian, Luxembourg, Italian equivalents… any format: .asc, .tif…), to be converted.
- Reduced EPSG / <Country> — the PREPARED files (converted to EPSG:4326 and reduced to 4/6/10 m), ready to assemble. Button shown: « EPSG 4326 · DEM-Alt (4/6/10 m) ».
- Sonny data / <Country> — the Sonny relief, kept separate (licence). Used as a FALLBACK when a country has no native data available.
- Tile assembly / <tile number> — the final tile <tile>.tif, the one feeding custom_dem.

Step-by-step workflow

1) Load the data (from the providers)

Download the RAW elevation data from the official organisations, country by country. Each provider splits its data differently:

- France (IGN, RGE ALTI): one file per DEPARTMENT.
- Belgium: by region, via THREE separate providers — Wallonia, Flanders, Brussels.
- Switzerland: ONE single official file for the whole country.
- Germany: no practical native source for now → fall back to the Sonny relief (see the Sonny relief section).

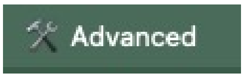
This data is « raw »: often in a national projection (e.g. Lambert-93 for France) and at a fine resolution (1 m). It will need to be converted (Step 3).

Put each download into the relevant country folder, inside « Source elevation » (see Step 2).

2) Open the module

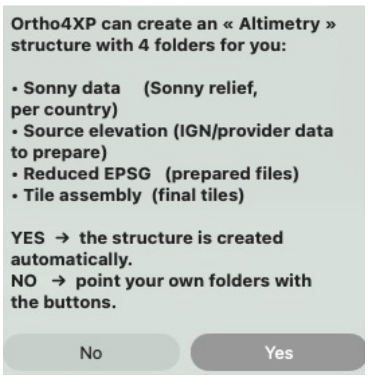
In ORTHO4XP_V3 → Advanced menu → Altimetry / DEM.
The window opens with the current tile in its title (example: +48+007).
[Screen: « Altimetry / DEM — +48+007 » window. Top: the step reminder; centre: the log (dark area); bottom: the buttons.]

Click « Advanced ».



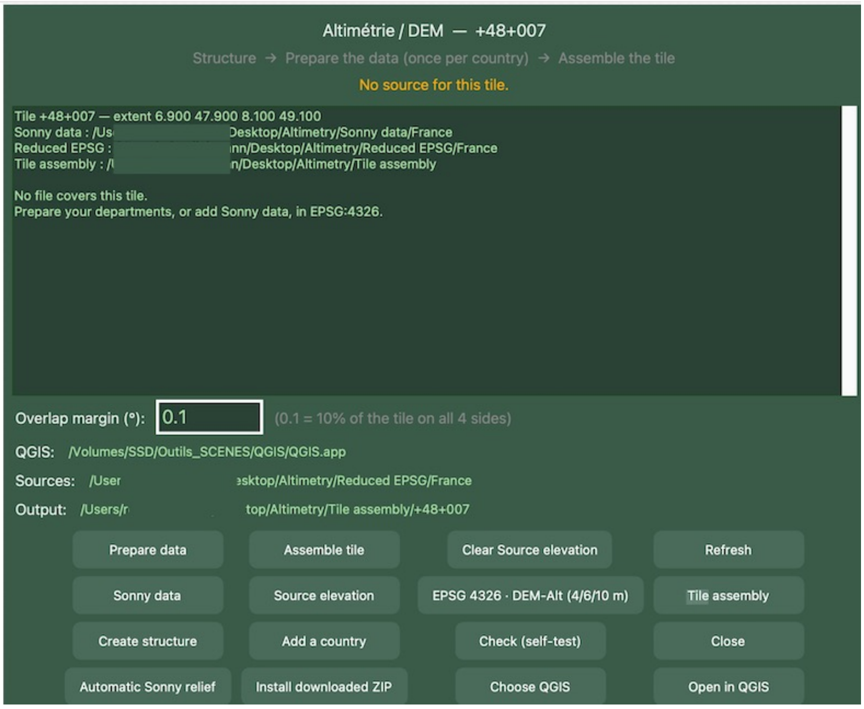
Click « Altimetry/DEM/QGIS ».

The first time, if the structure does not exist, ORTHO4XP_V3 offers to create it. Click Yes.



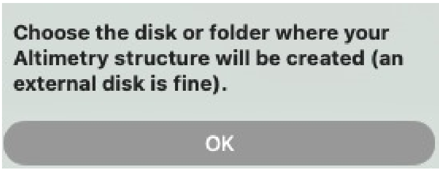
2) Set up the structure

1) Structure



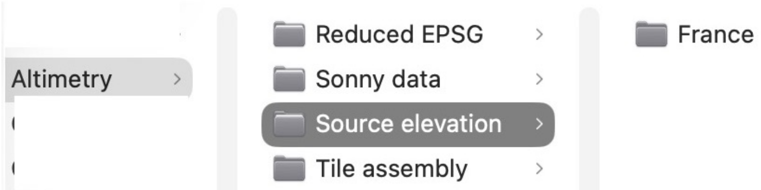
Click « Create structure ».

Choose the host disk or folder (an external disk is fine). Avoid Desktop and Documents (see the macOS warning below).



Enter the name of a first country (e.g. France).

The module creates all 4 folders at once, with the country sub-folder in the first 3.



For each additional country (Switzerland, Belgium, Germany...), click « Add a country » and enter its name. The sub-folder is created in Source elevation, Reduced EPSG and Sonny data — never in Tile assembly (which is organised by tile number).

You can create your own sub-sub-folders in Finder (e.g. Source elevation / Belgium / Wallonia): the scan goes down recursively, so you organise them freely.

> *IMPORTANT: the folder buttons (Source elevation, Reduced EPSG, Sonny data, Tile assembly) only OPEN an existing folder. They never create the tree: only « Create structure » does.*

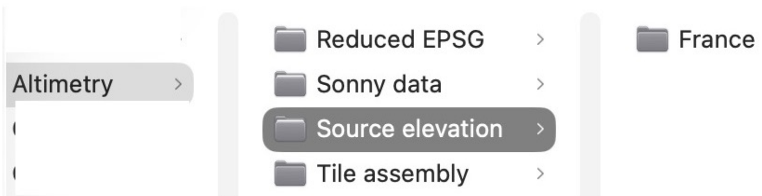
3) Prepare the data

1) Prepare the data (convert + reduce)

To be done once per country / department. Reproduces the QGIS chain gdalbuildvrt → gdalwarp (to EPSG:4326) → gdal_translate (reduction).

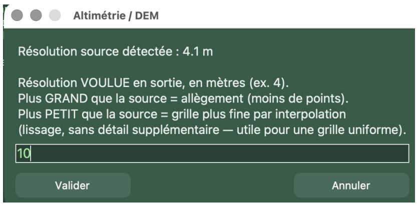
Click « Prepare data ».

Choose the raw-data folder to prepare (it opens on Source elevation): for example Source elevation / France.



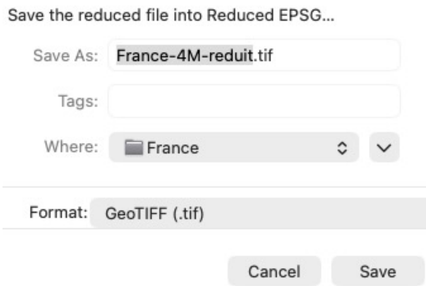
> *The module detects the source resolution (e.g. 1 m for IGN, ~27 m for Sonny).*

Enter the desired output resolution, in metres (e.g. 10), and confirm.

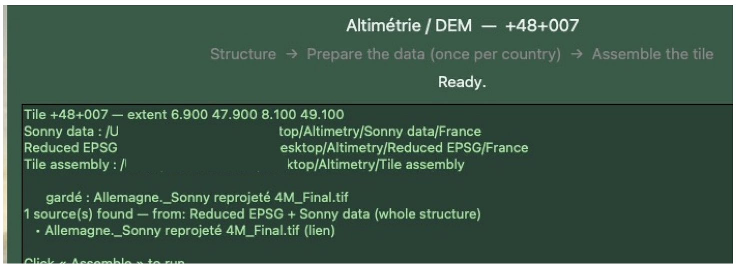


- ➢ Larger than the source → lightening (fewer points).
- ➢ Finer than the source → interpolation (denser grid, useful to standardise all sources to 10 m before assembly).

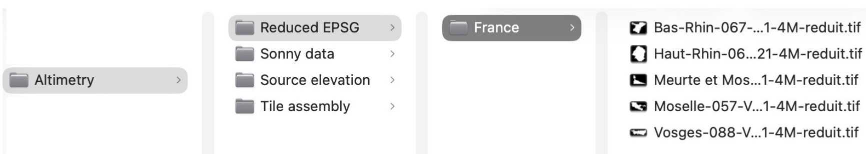
Reduced EPSG is selected by default. Save the file there (e.g. France-10M-reduit.tif).



Wait until processing finishes.



Result: a file already in EPSG:4326, reduced, stored in Reduced EPSG, ready for assembly.



> *Sonny note — prepare it or not? It depends on the resolution of the OTHER data on the tile:*

- ➢ If your other elevation data is coarse (20–30 m): use Sonny AS IS, without preparing it — it is already at the same level.

➢ If your other elevation data is fine (4 m, 6 m or 10 m): adapt Sonny (20–30 m) to THAT same resolution via « Prepare », only to avoid « cliff » effects (steps) at the joints in X-Plane.

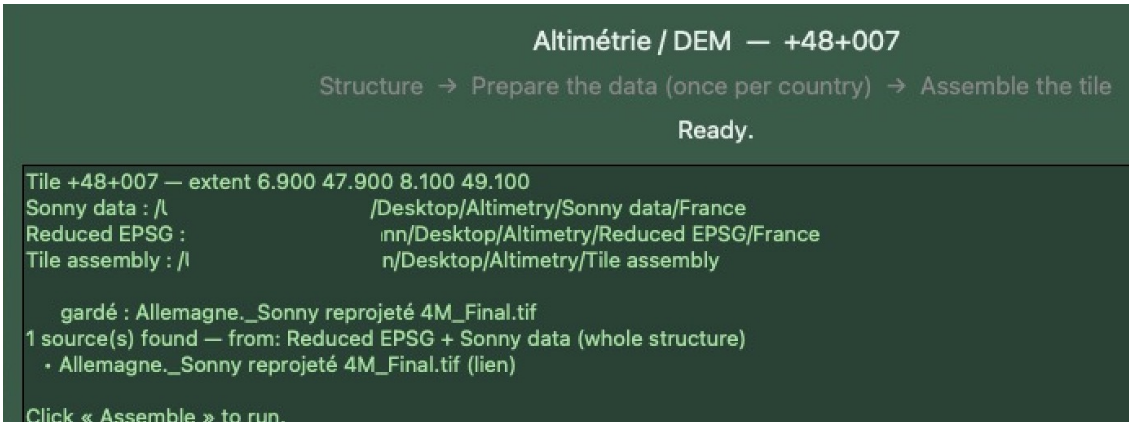
⚠️ *In all cases you DO NOT GAIN precision or detail: Sonny stays interpolated 20–30 m data. It is simply aligned onto the neighbouring data grid. Sonny is already in EPSG:4326.*

4) Assemble the data

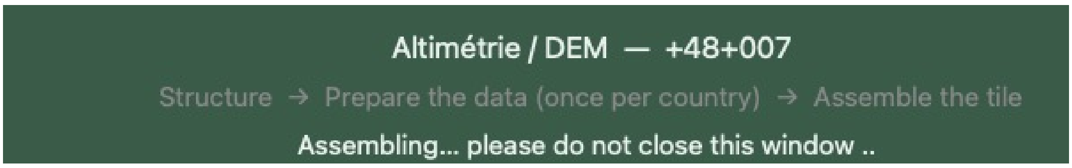
1) Assemble the tile

By default the active tile in ORTHO4XP_V3 is the one entered in the Latitude/Longitude interface (e.g. +48+7).

Optionally click « Refresh »: the log lists the sources found (kept / skipped + reason).



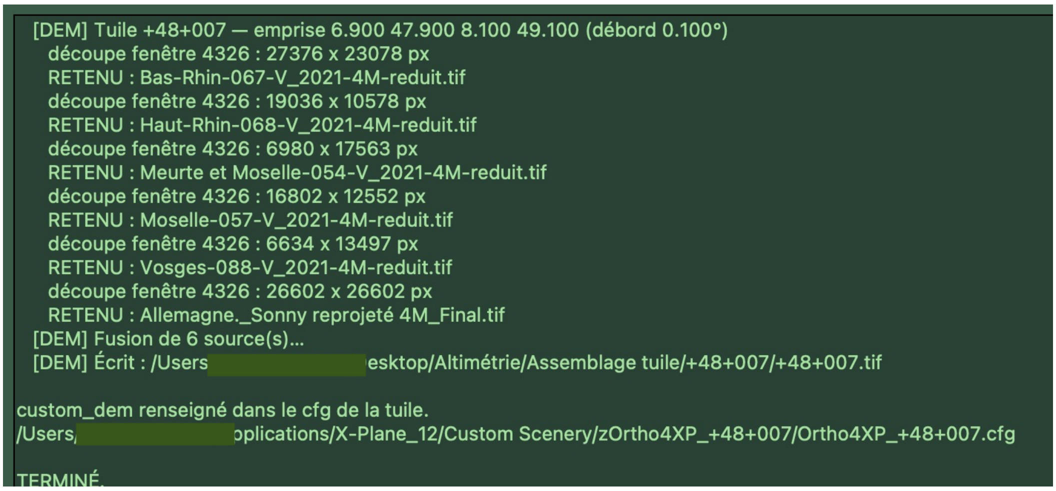
Click « Assemble tile ».



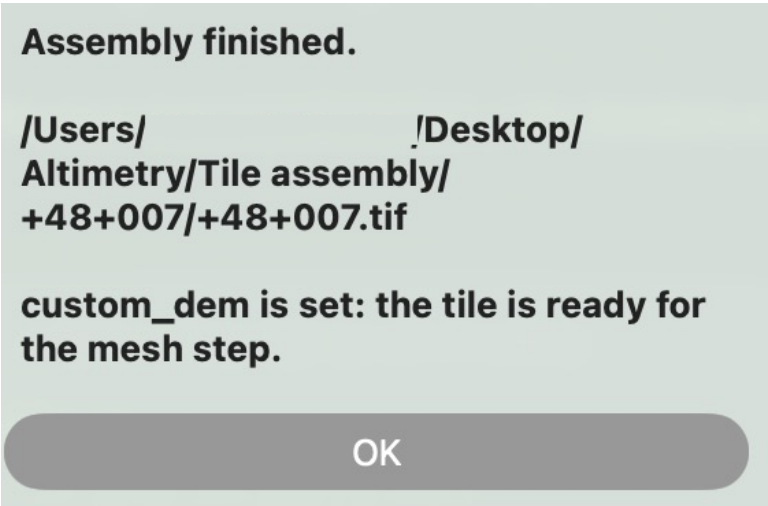
The module scans the WHOLE structure — all of Reduced EPSG and all of Sonny data, every country — and keeps only the files covering the tile enlarged by the overlap. It crops them, merges them, writes Tile assembly / <tile> / <tile>.tif, then fills in custom_dem in the tile's .cfg.

Real example (+48+007, Alsace): 5 French departments (Reduced EPSG) + the Germany relief (Sonny data) are selected and merged automatically, with no manual tuning.

[Screen: the log lets you follow the steps « window crop 4326 … / KEPT … » for each source, then « Merging N source(s)… » and « Written: …/<tile>.tif ».]



Operation complete



5) Very large country files

A file covering a whole country can weigh tens of GB (e.g. Germany 4M ≈ 148 GB). The module NEVER loads it entirely: if it is already in EPSG:4326, it crops only the tile WINDOW (read in row blocks, constant memory) — the equivalent of gdalwarp -te. Assembly stays fast whatever the size.

> *Tip: to avoid duplicating a 148 GB file, place a SYMBOLIC LINK to the real file in Sonny data / <Country>. The module follows the link, even if its name does not end in .tif.*

6) Clear Source elevation (cleanup)

Button: Clear Source elevation

Once your departments are converted, you can free up space. The button shows a CHECKLIST of the departments present, with an indicator: ✅ already reduced (safe to delete) / ⚠️ not reduced yet (keep it).

Triple safety: nothing is deleted without (1) your tick, (2) a warning if a non-reduced department is ticked, (3) a final confirmation. Never automatic, never in bulk.

7) Sonny relief (fallback)

Buttons: Automatic Sonny relief / Install downloaded ZIP

When a country has no native data, use the Sonny relief. « Automatic Sonny relief » detects whether the tile is covered and, if not, opens the download site. « Install downloaded ZIP » unpacks the .hgt directly into Sonny data / <Country>, checks it is in place, then offers to delete the .zip (never automatically).
No separate disk to define any more.

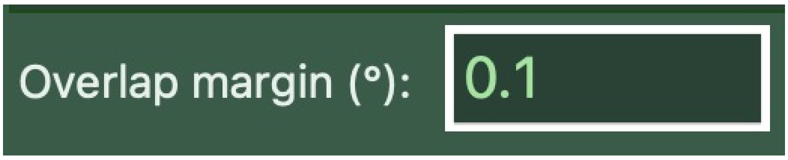
Sonny resolution and « cliff » effect: Sonny is reworked relief, ~20–30 m resolution, less precise than native data (IGN, etc.). Two cases:

- Other tile data at 20–30 m → Sonny usable as is.
- Other data at 4 / 6 / 10 m → prepare Sonny to the same resolution (4, 6 or 10 m) to avoid steps at the joints in X-Plane.

The goal is only to harmonise the grid to remove « cliff » effects between a Sonny area and a native area — not to improve precision, which stays that of the original Sonny.

8) Permanent settings

Overlap (°)



Default value 0.1 (= 10 % of the 1° tile on all 4 sides). Ensures neighbouring tile edges overlap seamlessly. Do not change unless needed.

QGIS / Choose QGIS / Open in QGIS

Optional path to the QGIS application, to open the assembled <tile>.tif and check the relief.

9) Result and verification

The <tile>.tif file is written in Tile assembly / <tile> and pointed to automatically by custom_dem. Check in QGIS that the tile is FULL (statistic VALID_PERCENT = 100, no gaps), then run the build in
X-Plane 12 to check the relief in flight.

> *Note: the produced file is compressed (DEFLATE) — so it is lighter than an uncompressed QGIS export, with strictly identical data.*

macOS warning — permissions

If the log shows « Operation not permitted » on a file, it is a macOS privacy protection: your structure is probably on the Desktop or in Documents. Fixes: allow the application in System Settings → Privacy & Security → Full Disk Access, OR move the Altimetry structure out of Desktop / Documents.